

Gaussian Algebra

The *Gaussian* or *normal* probability distribution over $\mathbb{R}^D$ is identified by its probability density function

$$\mathcal{N}(x; \mu, \Sigma) = \frac{1}{(2\pi)^{D/2} |\Sigma|^{1/2}} \exp \left(-\frac{1}{2} (x - \mu)^\top \Sigma^{-1} (x - \mu) \right). \quad (3.1)$$

Here, a parameter vector $\mu \in \mathbb{R}^D$ specifies the *mean* of the distribution, and a *symmetric positive definite* (SPD) matrix $\Sigma \in \mathbb{R}^{D \times D}$ defines the *covariance* of the distribution¹

$$\mu_i = \mathbb{E}_{\mathcal{N}(x; \mu, \Sigma)}(x_i), \quad \Sigma_{ij} = \text{cov}_{\mathcal{N}(x; \mu, \Sigma)}(x_i, x_j).$$

There are many ways to motivate the prevalence of the Gaussian distribution. It is sometimes presented as arising from analytic results like the central limit theorem, or the fact that the normal distribution is the unique probability distribution with mean μ and covariance Σ maximising the differential entropy functional.² But the primary practical reason for the ubiquity of Gaussian probability distributions is that they have convenient algebraic properties. This is analogous to the popularity of *linear* approximations in numerical computations: The main reason to construct linear approximations is that linear functions offer a rich analytic theory, and that computers are good at the basic linear operations – addition and multiplication.

In fact, the connection between linear functions and Gaussian distributions runs deeper: Gaussians are a family of probability distributions that are preserved under all linear operations. The following properties will be used extensively:

- ⊖ If a variable $x \in \mathbb{R}^D$ is normal distributed, then *every affine transformation* of it also has a Gaussian distribution (Fig-

¹ The matrix inverse Σ^{-1} is known as the *precision* matrix. The meaning of its elements is a bit more involved than for those of Σ , and pertains to the distribution of one or two variables *conditional* on all the others: If $p(x) = \mathcal{N}(x; \mu, \Sigma)$, then the variance of x_i conditioned on the value of $x_j, j \neq i$ is

$$\text{var}_{|x_{j \neq i}}(x_i) = 1/[\Sigma^{-1}]_{ii}.$$

An interpretation for the off-diagonal elements of Σ^{-1} is that they provide the coefficients for a linear equation that determines the expectation of one element of the variable when conditioned on all the others (cf. Eq. (3.10)). Assume $\mu = 0$ for simplicity. Then, given the information $x_{j \neq i} = y$, the expectation of x_i is

$$\mathbb{E}_{|x_{j \neq i}=y}(x_i) = -\frac{1}{[\Sigma^{-1}]_{ii}} \sum_{j \neq i} [\Sigma^{-1}]_{ij} y_j.$$

More on this in a report by MacKay (2006).

² The entropy,

$$\mathbb{H}_p(x) := -\int p(x) \log p(x) dx, \quad (3.2)$$

of the Gaussian is given by

$$\begin{aligned} \mathbb{H}_{\mathcal{N}(x; \mu, \Sigma)}(x) \\ = \frac{N}{2} (1 + \log(2\pi)) + \frac{1}{2} \log |\Sigma|. \end{aligned} \quad (3.3)$$

ure 3.1):

$$\begin{aligned} &\text{if } p(\mathbf{x}) = \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}, \boldsymbol{\Sigma}), \\ &\quad \text{and } \mathbf{y} := \mathbf{Ax} + \mathbf{b} \text{ for } \mathbf{A} \in \mathbb{R}^{M \times D}, \mathbf{b} \in \mathbb{R}^M, \\ &\text{then } p(\mathbf{y}) = \mathcal{N}(\mathbf{y}; \mathbf{A}\boldsymbol{\mu} + \mathbf{b}, \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^\top). \end{aligned} \quad (3.4)$$

- ⊕ The product of two Gaussian probability density functions is another Gaussian probability distribution, scaled by a constant.³ The value of that constant is itself given by the value of a Gaussian density function

$$\begin{aligned} \mathcal{N}(\mathbf{x}; \mathbf{a}, \mathbf{A})\mathcal{N}(\mathbf{x}; \mathbf{b}, \mathbf{B}) &= \mathcal{N}(\mathbf{x}; \mathbf{c}, \mathbf{C})\mathcal{N}(\mathbf{a}; \mathbf{b}, \mathbf{A} + \mathbf{B}), \\ \text{where } \mathbf{C} &:= (\mathbf{A}^{-1} + \mathbf{B}^{-1})^{-1}, \\ \text{and } \mathbf{c} &:= \mathbf{C}(\mathbf{A}^{-1}\mathbf{a} + \mathbf{B}^{-1}\mathbf{b}). \end{aligned} \quad (3.5)$$

These two properties also provide the mechanism for *Gaussian inference*: If the variable $\mathbf{x} \in \mathbb{R}^D$ is assigned a Gaussian prior, and observations $\mathbf{y} \in \mathbb{R}^M$, given $\mathbf{x}$, are Gaussian distributed

$$p(\mathbf{x}) = \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}, \boldsymbol{\Sigma}) \quad \text{and} \quad p(\mathbf{y} | \mathbf{x}) = \mathcal{N}(\mathbf{y}; \mathbf{Ax} + \mathbf{b}, \boldsymbol{\Lambda}),$$

then both the posterior and the marginal distribution for $\mathbf{y}$ (the evidence) are Gaussian (Figure 3.2):

$$p(\mathbf{x} | \mathbf{y}) = \mathcal{N}(\mathbf{x}; \tilde{\boldsymbol{\mu}}, \tilde{\boldsymbol{\Sigma}}), \quad \text{with} \quad (3.6)$$

$$\tilde{\boldsymbol{\Sigma}} := (\boldsymbol{\Sigma}^{-1} + \mathbf{A}^\top \boldsymbol{\Lambda}^{-1} \mathbf{A})^{-1} \quad (3.7)$$

$$= \boldsymbol{\Sigma} - \boldsymbol{\Sigma} \mathbf{A}^\top (\mathbf{A} \boldsymbol{\Sigma} \mathbf{A}^\top + \boldsymbol{\Lambda})^{-1} \mathbf{A} \boldsymbol{\Sigma}, \quad \text{and} \quad (3.8)$$

$$\tilde{\boldsymbol{\mu}} := \tilde{\boldsymbol{\Sigma}} (\mathbf{A}^\top \boldsymbol{\Lambda}^{-1} (\mathbf{y} - \mathbf{b}) + \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}) \quad (3.9)$$

$$= \boldsymbol{\mu} + \boldsymbol{\Sigma} \mathbf{A}^\top (\mathbf{A} \boldsymbol{\Sigma} \mathbf{A}^\top + \boldsymbol{\Lambda})^{-1} (\mathbf{y} - (\mathbf{A}\boldsymbol{\mu} + \mathbf{b})); \quad (3.10)$$

and

$$p(\mathbf{y}) = \mathcal{N}(\mathbf{y}; \mathbf{A}\boldsymbol{\mu} + \mathbf{b}, \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^\top + \boldsymbol{\Lambda}). \quad (3.11)$$

The equivalent forms, Eqs. (3.9) & (3.10) and Eqs. (3.7) & (3.8), show two different formulations of the same posterior mean and covariance, respectively. The former pair contains a matrix inverse of size $D \times D$, the latter one of size $M \times M$. Depending on which of the two numbers is larger, it is more efficient to compute one or the other. The marginal covariance of $\mathbf{y}$ in Eq. (3.11) also shows up as part of Eq. (3.8). Computing this evidence term thus adds only minimal computational overhead over the computation of the posterior mean.

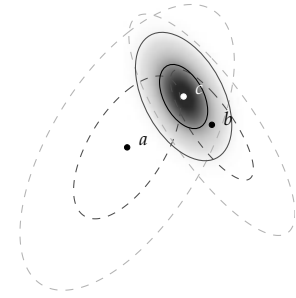


Figure 3.1: The product of two Gaussian densities is another Gaussian density, up to normalisation.

³ This statement is about the product of two probability density functions. In contrast, the product of two Gaussian random variables is *not* a Gaussian random variable.

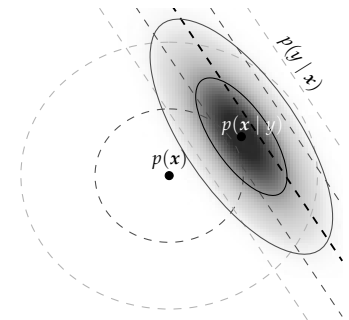


Figure 3.2: The posterior distribution $p(\mathbf{x} | \mathbf{y})$ arising from a Gaussian prior

$$p(\mathbf{x}) = \mathcal{N}(\mathbf{x}; [1, 0.5]^\top, 3^2 \mathbf{I})$$

and Gaussian likelihood

$$p(y = 6 | \mathbf{x}) = \mathcal{N}(6; [1, 0.6]^\top \mathbf{x}, 1.5^2)$$

is itself Gaussian. This sketch also illustrates that the likelihood function does not need to be a proper probability distribution on $\mathbf{x}$ (only on $\mathbf{y}$). The posterior distribution can be correlated, even if the prior is uncorrelated. This phenomenon, that two a priori independent “parent” variables x_1, x_2 can become correlated by an observation $y = a_1 x_1 + a_2 x_2$ connecting the two, is known as *explaining away* (Pearl, 1988).

An important special case arises if the matrix A maps to a proper subset of x . Consider a separation of $x = [a, b]^T$ into $a \in \mathbb{R}^d$ and $b \in \mathbb{R}^{D-d}$. The joint is⁴

$$p(x) = \mathcal{N} \left(\begin{bmatrix} a \\ b \end{bmatrix}; \begin{bmatrix} \mu_a \\ \mu_b \end{bmatrix}, \begin{bmatrix} \Sigma_{aa} & \Sigma_{ab} \\ \Sigma_{ba} & \Sigma_{bb} \end{bmatrix} \right).$$

Now consider $A = [I_d, \mathbf{0}_{D-d}]$, i.e. the “selector” map extracting a subset of size $d \leq D$. By Eq. (3.4), the *marginal* of this Gaussian distribution is another Gaussian, whose mean and covariance are simply a sub-vector and sub-matrix of the full mean and covariance, respectively (Figure 3.3)

$$p(a) = \int p(a, b) db = \mathcal{N}(a; \mu_a, \Sigma_{aa}). \quad (3.12)$$

Additionally using Eq. (3.6), we see that the *conditional* of a subset conditioned on its complement is also a Gaussian:

$$p(a | b) = \mathcal{N}(a; \mu_a + \Sigma_{ab}(\Sigma_{bb})^{-1}b, \Sigma_{aa} - \Sigma_{ab}(\Sigma_{bb})^{-1}\Sigma_{ba}). \quad (3.13)$$

Since marginalisation (sum rule) and conditioning (product rule) are the two elementary operations of probability theory, “Gaussian distributions map probability theory to linear algebra” – to matrix multiplication and inversion.

⁴ In other words, if we only care about a small number d of a large set of D interdependent (co-varying) Gaussian variables, then that marginal can be computed trivially, by selecting elements from the mean vector and covariance matrix. On the one hand, this property shows that covariance only captures a limited amount of structure. On the other, this makes it possible to consider arbitrarily large sets of variables, as long as we only have to deal with finite subsets of them. This observation is at the heart of the notion of a Gaussian process, discussed below.

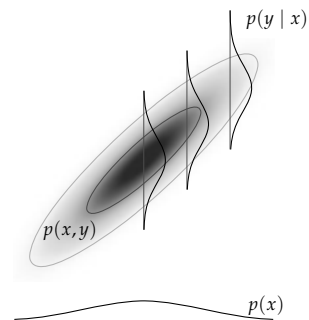


Figure 3.3: Projections and conditionals of a Gaussian are Gaussians.

